

CHARACTERIZATION OF THE EXPRESSION OF INDUCIBLE NITRIC OXIDE SYNTHASE IN RAT AND HUMAN LIVER DURING HEMORRHAGIC SHOCK

Joy L. Collins,* Yoram Vodovotz,* Christian Hierholzer,*
Raphael T. Villavicencio,* Shubing Liu,* Sean Alber,[†] David Gallo,*
Donna B. Stolz,[†] Simon C. Watkins,[†] Anthony Godfrey,* William Gooding,[‡]
Edward Kelly,* Andrew B. Peitzman,* and Timothy R. Billiar*

Departments of *Surgery and [†]Cell Biology and Physiology, and [‡]Biostatistics Facility, University of Pittsburgh Cancer Institute, University of Pittsburgh, Pittsburgh, Pennsylvania 15213

Received 25 Apr 2002; first review completed 10 May 2002; accepted in final form 7 Aug 2002

ABSTRACT—It has been previously shown that the inducible nitric oxide (NO) synthase (iNOS; NOS-2) is elevated after hemorrhage, and that iNOS-derived NO participates in the upregulation of inflammation as well as lung and liver injury postresuscitation from shock. The purpose of this study was to elucidate the time course of iNOS mRNA expression, as well as the cellular and subcellular localization of iNOS protein in the liver posthemorrhage in rats subjected to varying durations of hemorrhagic shock (HS; mean arterial blood pressure [MAP] = 40 mmHg) with or without resuscitation. Expression of iNOS mRNA in rat liver by real-time reverse transcriptase (RT)-PCR demonstrated iNOS upregulation in shocked animals as compared with their sham counterparts as early as 60 min after the initiation of hemorrhage. By 1 h of HS, iNOS protein was detectable in rat liver by immunofluorescence, and this expression increased with time. Immunofluorescence localized iNOS primarily to the hepatocytes, and in particular to hepatocytes in the centrilobular regions. This analysis, confirmed by immunoelectron microscopy, revealed that iNOS colocalizes with catalase, a peroxisomal marker. Furthermore, we determined that iNOS mRNA is detectable by RT-PCR in liver biopsies from human subjects with HS (MAP < 90 mmHg) associated with trauma (n = 18). In contrast, none of the seven nontrauma surgical patients studied had detectable iNOS mRNA in their livers. Collectively, these results suggest that hepatic iNOS expression, associated with peroxisomal localization, is an early molecular response to HS in experimental animals and possibly in human patients with trauma with HS.

KEYWORDS—Hemorrhage, hemorrhagic shock, nitric oxide, iNOS, rat, real-time PCR

INTRODUCTION

Hemorrhagic shock (HS) initiates a systemic inflammatory response involving the upregulation of cytokine expression and accumulation of neutrophils. One consequence of such changes is the potential for end organ dysfunction and damage, which are prominent in the lungs and liver. A delayed consequence of the initial hyperinflammation is a counter anti-inflammatory response that renders the host more susceptible to delayed infection, sepsis, and multiple organ failure (1, 2). Inducible nitric oxide (NO) synthase (iNOS; NOS-2) is elevated after hemorrhage (3–5). Moreover, iNOS-derived NO participates in the upregulation of inflammation immediately postresuscitation from shock and is associated with shock-induced lung and liver injury (5). Inhibition of iNOS with the selective inhibitor N⁶-(iminoethyl)-L-lysine (L-NIL) resulted in a marked reduction of liver injury following HS. Animals deficient in the gene for iNOS also demonstrated reduced lung and liver damage after HS when compared with their wild-type counterparts (5). A prominent feature of this study was the observation that the absence or suppression of iNOS resulted in a marked decrease in NF- κ B activation, cytokine mRNA expression, and neutrophil influx postresuscitation (5).

The observations described above suggest that the upregu-

lation of iNOS during HS may occur rapidly. To assess the expression pattern of iNOS during HS, we examined the temporal characteristics of iNOS mRNA and protein in the liver after the initiation of HS. This analysis was coupled to experiments to define the cellular and subcellular distribution of the enzyme with the liver. To provide evidence for clinical relevance, we also tested for the expression of iNOS mRNA in liver biopsies from patients undergoing emergency laparotomy post-trauma. We demonstrate that iNOS mRNA is elevated by 60 min of shock and that iNOS protein is localized primarily to hepatocytes. Within hepatocytes, the protein is detected in both the cytosol and peroxisomes.

MATERIALS AND METHODS

Human tissue samples

This protocol was in accordance with the National Institutes of Health guidelines for human clinical studies and was approved by the Institutional Review Board of the University of Pittsburgh. Tru-cut liver biopsies were taken from 18 patients with trauma presenting to the University of Pittsburgh Medical Center from January 1, 1995 to January 10, 1996 who underwent urgent laparotomy for control of acute hemorrhage and abdominal injuries. All biopsies were performed within 3 h of presentation to the emergency room and were immediately frozen in liquid nitrogen and stored at –80°C until analysis. Sections from hepatic lobes resected electively from patients without trauma for metastatic disease served as controls. Total RNA was extracted from all specimens using the guanidinium isothiocyanate method (6), and cDNA templates were synthesized by the reverse transcriptase (RT) reaction.

Hemorrhagic shock protocol

This study was approved by the University of Pittsburgh Institutional Animal Care and Use Committee and conformed to National Institutes of Health guidelines for the care and use of laboratory animals. Male Sprague-Dawley rats (250–300 g)

Address reprint requests to Timothy Billiar, MD, Department of Surgery, University of Pittsburgh, F-1281 Presbyterian University Hospital, Pittsburgh, PA 15213.

DOI: 10.1097/01.shk.0000046093.26538.2a

were obtained from Harlan (Indianapolis, IN). Fasted animals were anesthetized with a single intraperitoneal injection of pentobarbital (45 mg/kg). The animals were then restrained and a midline neck incision was made. The rats were then orotracheally intubated with a 14-gauge plastic Jelco catheter (Johnson & Johnson, Arlington, TX) for the purpose of maintaining a secure airway. The right carotid artery was cannulated for continuous blood pressure monitoring and blood withdrawal, and the left jugular vein was cannulated for fluid administration. After vascular cannulation, blood was aspirated into a heparinized syringe over a period of 15 min. This initial blood withdrawal brought the mean arterial pressure (MAP) from a baseline of approximately 120 mmHg down to 40 mmHg, thereby establishing moderate to severe shock. Total bleed-out volume was 7–9 mL. After this initial hemorrhage, blood was withdrawn and returned as needed to maintain a MAP of 40 mmHg for the duration of the shock period. Animals then underwent either an HS protocol or an HS with resuscitation protocol.

In the HS protocol, animals were maintained at 40 mmHg and were sacrificed after 1, 2.5, 3.5, or 5 h. In the HS and resuscitation protocol, the animals underwent 2.5 h of shock, after which they received the remainder of their shed blood plus two shed blood volumes of lactated Ringer's solution via the jugular venous catheter. The resuscitation fluids were administered over approximately 45 min, after which the animals were allowed to emerge from anesthesia and were returned to their cages. The return of the remaining shed blood plus two times the shed blood volume of crystalloid brought the MAP to normal levels of 120–140 mmHg. Animals were sacrificed 4 h after the initiation of resuscitation.

In both the HS and HS with resuscitation protocols, sham control animals underwent anesthesia, cannulation, and restraint for an identical period of time as shock animals, but did not undergo hemorrhage. During the hypotensive period, animals in both HS and HS + resuscitation protocols received a continuous intravenous infusion of lactated Ringer's solution (pH 7.4) at a rate of 1 mL/h.

Isolation of liver and tissue processing

After completion of the respective protocols, animals were sacrificed by CO₂ asphyxiation and livers were removed. A portion of each liver was immediately cryopreserved in liquid nitrogen and was stored at –80°C until molecular analyses were performed. The remaining portion of liver tissue was placed in a solution of 2% paraformaldehyde in 1× phosphate-buffered saline (PBS) at 4°C for 6 h, followed by overnight storage in a 30% solution of sucrose in 1× PBS at 4°C. The fixed tissues were then cryopreserved in liquid nitrogen-cooled n-methylbutane and were stored at –80°C for later sectioning and immunohistochemical analysis.

RT-PCR analysis on human liver samples

Total RNA (2.5 µg) was subjected to first-strand cDNA synthesis using oligo (dT) primer and Moloney murine leukemia virus (MMLV) reverse transcriptase (Invitrogen-Gibco, Carlsbad, CA) as described elsewhere (7). Primers were designed to amplify rat β actin and iNOS with the assistance of a PCR primer design program (PCR Plan; Intelligenetics, Mountain View, CA). The sequences of the human and rat cytoplasmic β -actin 5' and 3' primers were TTCTACAATGAGC-TGCGTGTG and TTCATGGATGCCACAGGATTC, respectively. The sequence of the iNOS 5' primer was TTGGGTCTTGTAGCCTAGTC, corresponding to nucleotide numbers of the 114–134 bp of the rat iNOS cDNA sequence (Adachi 93). The sequence of the iNOS 3' primer was TGTGCAGTCCCAGTGAGGAAC, corresponding to nucleotide numbers 375–355. PCR conditions were as follows: denaturation at 94°C for 1 min, annealing at 57°C for 2 min, and polymerization at 72°C for 2 min. PCR reactions were performed in a DNA thermal cycler (Perkin-Elmer, Norwalk, CT) using different numbers of cycles to detect a linear range of input RNA. The optimal cycle number was identified as 30 cycles. The PCR products (5% of reaction volume) of qualitative RT-PCR reactions were separated electrophoretically on a 1.75% agarose gel and were stained with ethidium bromide.

Real-time RT-PCR analysis on rat liver samples

Quantitative RT-PCR was carried out on a 7700 Sequence Detector (Applied Biosystems, Foster City, CA) using the $\Delta\Delta C_T$ method of calculating relative expression as described previously (8–10). Duplicate RNA aliquots (100 and 400 ng) were reverse transcribed using random hexamers, and expression of the target genes was calculated relative to 18S rRNA as an internal control. All PCR reactions were carried out in duplicate on each RT reaction and the average $\Delta\Delta C_T$ was used to calculate relative expression. Reactions without reverse transcriptase (No-RT

controls) were run for each sample at the higher RNA input to ensure that signal detected was not from contaminating genomic DNA. The primers used are described in Table 1. Primer concentrations were 500 nM (iNOS) and 100 nM (18S).

Confocal imaging

Liver cryosections (5 µm) were collected onto gelatin-coated slides and were rinsed 3× in PBS, rinsed 3× in PBS containing 0.5% bovine serum albumin and 0.15% glycine (PBG), and were then blocked in 20% nonimmune goat serum in PBG for 30 min at room temperature. Primary antibodies, mouse anti-iNOS (Santa Cruz Biotechnology, Santa Cruz, CA, used at 1:50), and rabbit anti-catalase (Rockland, Gilbertsville, PA, used at 1:1500) were diluted in PBG and were incubated on sections for 60 min at room temperature. Sections were washed five times in PBG and then in secondary antibodies in PBG (goat anti-rabbit Cy3 used at 1:3000 [Jackson ImmunoResearch, West Grove, PA]) and goat anti-mouse Alexa 488 used at 1:500 (Molecular Probes, Eugene OR) were added to sections for 60 min at room temperature. Tissue was washed three times in PBG, three times in PBS, cover slipped using gelvatol [23 g of poly(vinyl) alcohol 2000, 50 mL of glycerol, and 0.1% sodium azide to 100 mL of PBS], and viewed on a confocal microscope (Leica, Malvern, PA).

Immunoelectron microscopy

Control, nonmanipulated liver and livers from rats hemorrhaged for 1, 2.5, 3.5, and 5 h were perfused with 2% paraformaldehyde and 0.01% glutaraldehyde in 0.1 M PBS and were stored at 4°C for 1 h. Pieces of the fixed liver tissue (1 mm³) were infused with 2.3 M sucrose in 0.1 M PBS overnight at 4°C. Tissue was frozen on ultracryotome stubs under liquid nitrogen and was stored in liquid nitrogen until use. Ultrathin sections (70–100 nm) were cut using a Reichert Ultracut U microtome with a FC4S cryoattachment, lifted on a small drop of 2.3 M sucrose, and mounted on Formvar-coated copper grids. Sections were washed three times with PBS, three times with PBS containing 0.5% bovine serum albumin, and 0.15% glycine (PBG), followed by a 30-min incubation with 5% normal goat serum in PBG. Sections were labeled with rabbit anti-bovine catalase (1:1500) and monoclonal anti-rat iNOS (1:50) in PBG for 1 h. Sections were washed four times in PBG and were labeled with goat anti-mouse (5 nm) and goat anti-rabbit (10 nm) gold-conjugated secondary antibodies (Amersham, Piscataway, NJ), each at a dilution of 1:25 for 1 h. Sections were washed three times in PBG, three times in PBS, and were then fixed in 2.5% glutaraldehyde in PBS for 5 min, washed two times in PBS, then washed six times in sterile water. Sections were poststained in 2% neutral uranyl acetate for 7 min, washed three times in sterile water, stained for 2 min in 4% uranyl acetate, and then embedded in 1.25% methyl cellulose. Labeling was observed on an electron microscope (JEM 1210; JEOL, Peabody, MA) at 80 kV.

Statistics

Unless otherwise indicated, data are presented as mean $\pm$ SEM. Comparisons of means were performed using analysis of variance (ANOVA). For analysis of real-time RT-PCR results, the following procedure was used. Relative iNOS mRNA expression was transformed to a distribution that was more Gaussian and more homoscedastic. The transformation selected was a back transform to $\Delta\Delta C_T$ by taking the negative of the base 2 logarithm of iNOS mRNA expression. The experimental design was a two-way, completely randomized factorial design. The factors were treatment group (shock or control) and duration of bleeding/anesthetization (60, 90, and 150 min, and 150 min with resuscitation at 4 h). Between five and six rats were assigned to each of the eight groups formed by crossing treatment and duration. Analysis of data was conducted with a two-way ANOVA. Treatment by duration interaction was tested and considered significant, prompting further tests of treatment group differences by duration and duration differences by treatment group. In addition, orthogonal polynomial contrasts were tested within the treatment groups. Reported *P* values pertain to *F* or two-tailed *t* tests for the transformed data.

RESULTS

iNOS mRNA and protein expression is increased in the livers of shocked rats

Others have demonstrated that iNOS activity is increased in several tissues, including the liver, following HS alone (3).

TABLE 1. Real-time RT-PCR primers

Primer type	iNOS	18S
Forward primer	CTTTACGCCACTAACAGTGGCA	CCCTGTAATTGGAATGAGTCCAC
Reverse primer	AGTCATGCTTCCCACGCTC	GCTGGAATTACCGCGGCT
Quencher probe	CACCACAAGGCCACCTCGGATATCTC	TGCTGGCACCAGCACTTGCCCTC

Furthermore, we have shown that inflammation following HS is mediated by iNOS (5). However, neither the time required for iNOS to be first upregulated nor the cellular distribution of iNOS in the liver have been established. We first carried out a time course comparing iNOS mRNA levels in shocked rats, with and without subsequent resuscitation, with levels in sham-shocked rats and in normal control rats. Real-time RT-PCR analysis demonstrated a pattern of increased iNOS mRNA expression in the livers of hemorrhaged animals as compared with time-matched sham animals within 90 min of the initiation of hemorrhage, though iNOS was present by 60 min in both sham-treated and shocked animals (Fig. 1). The elevation in iNOS mRNA expression in shocked animals was marginally significant ($P = 0.060$) at 90 min and highly significant at 150 min of HS with resuscitation at 4 h ($P = 0.020$), but not at 60 or 90 min.

iNOS protein in the livers of shocked rats

To characterize the cellular and subcellular expression of iNOS, immunostaining for iNOS was performed on liver sections from shocked ($n = 3$) and sham-shocked ($n = 3$) rats. In sham rat livers, iNOS staining was absent or minimal. In the livers of shocked animals, however, iNOS staining was evident; interestingly, a punctate immunostaining pattern of iNOS protein was observed (Fig. 2). The staining was localized to hepatocytes rather than to other cell types within the liver. The most intense staining was seen in the hepatocytes in the centrilobular region, surrounding the central veins (CV). Importantly, protein expression was evident by 1 h and increased with time using this analysis (Fig. 2). We have previously shown in endotoxemia that iNOS localizes to peroxisomes in hepatocytes (11). To establish if iNOS expression was peroxisomal in HS, we determined if iNOS is colocalized with catalase, a prominent constituent of peroxisomes. The

bottom panels of Figure 2 demonstrate iNOS staining in sections from animals that underwent 5 h of HS. The right panels show catalase staining, the left panels show iNOS staining, and the merged images in the central panel demonstrate a significant amount of colocalization of iNOS and catalase within the same cells. It is important to note that not all cells expressed iNOS protein, but where iNOS was present, it often colocalized with catalase. Western blot analysis with anti-iNOS antibody of whole liver from rats subjected to HS for 5 h confirmed the presence of a ~135 kD band (data not shown).

iNOS colocalizes with catalase in hepatocytes of shocked rats

The punctate nature of the iNOS staining suggested that iNOS is contained within, and not simply associated with, the external portion of the peroxisome. To examine the suborganellar localization of iNOS in peroxisomes at higher resolution, immunogold labeling and electron microscopy were performed (Fig. 3). Anti-iNOS immunoreactivity was broadly distributed in association with structures having the characteristic appearance of peroxisomes, as determined by the presence of the urate oxidase crystalline core of the peroxisome (Fig. 3). This analysis suggested that iNOS is found in the matrix of the peroxisome, and not in the core, which consists mostly of urate oxidase crystals. It should also be noted that iNOS immunoreactivity was not detected in preferential association with the plasma membrane, nucleus, nuclear lamina, endoplasmic reticulum, or mitochondria.

iNOS expression is increased in the livers of human patients with trauma

To establish the clinical relevance of iNOS upregulation following HS (MAP < 90 mmHg) associated with trauma, we analyzed tissues from liver biopsies in hypotensive patients with trauma and in surgical patients without trauma for the presence of iNOS mRNA. Liver biopsies were obtained from a total of 25 patients. Eighteen of these were patients with trauma who were hypotensive (MAP < 90 mmHg) for a period of 60–120 min before undergoing emergency abdominal laparotomy. Seven of these patients were patients without trauma undergoing elective liver resection for metastatic disease, and therefore served as controls. RT-PCR analysis clearly demonstrated that iNOS mRNA was present in all samples from hypotensive patients with trauma, but was not detected in the liver samples of patients without trauma (Fig. 4).

DISCUSSION

This study was undertaken to characterize the time course and distribution of iNOS expression during HS. Our findings indicate that hepatic iNOS expression is an early molecular response to controlled HS in experimental animals, and also characterizes the early response of humans subjected to trauma. We have extended previous studies of rodent HS (3, 5, 12) by showing that iNOS mRNA is upregulated early during HS (within 60–90 min by real-time RT-PCR) and by immunofluorescence for iNOS protein by 1 h. The upregulation of iNOS during HS is consistent with the findings of Hierholzer et al. (5) and Menezes et al. (13, 14), which demonstrate that

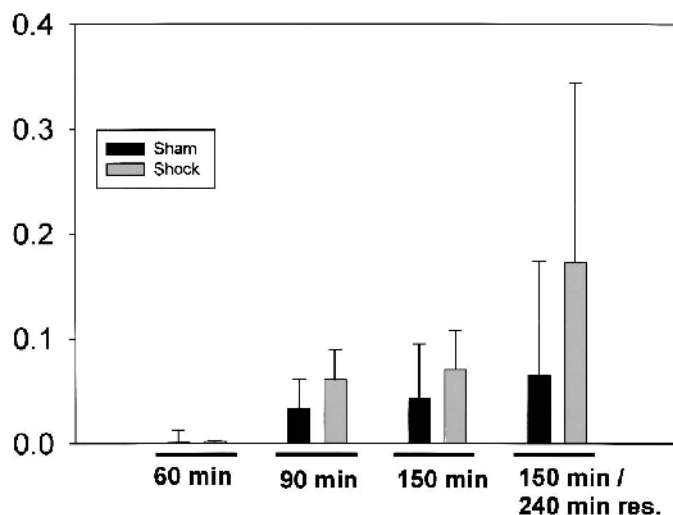


FIG. 1. Real-time RT-PCR analysis of iNOS mRNA expression in livers of rats subjected to sham shock and hemorrhagic shock with and without resuscitation. Rats were subjected to HS alone or with resuscitation ("res."), or sham procedure, as indicated. Livers were obtained and total RNA was prepared and subjected to analysis of iNOS mRNA using real time RT-PCR analysis as described in "Materials and Methods." Bars represent the relative iNOS expression in hemorrhaged animals and sham animals at the corresponding time point. Error bars are group standard deviations ($n = 4$ animals per group).

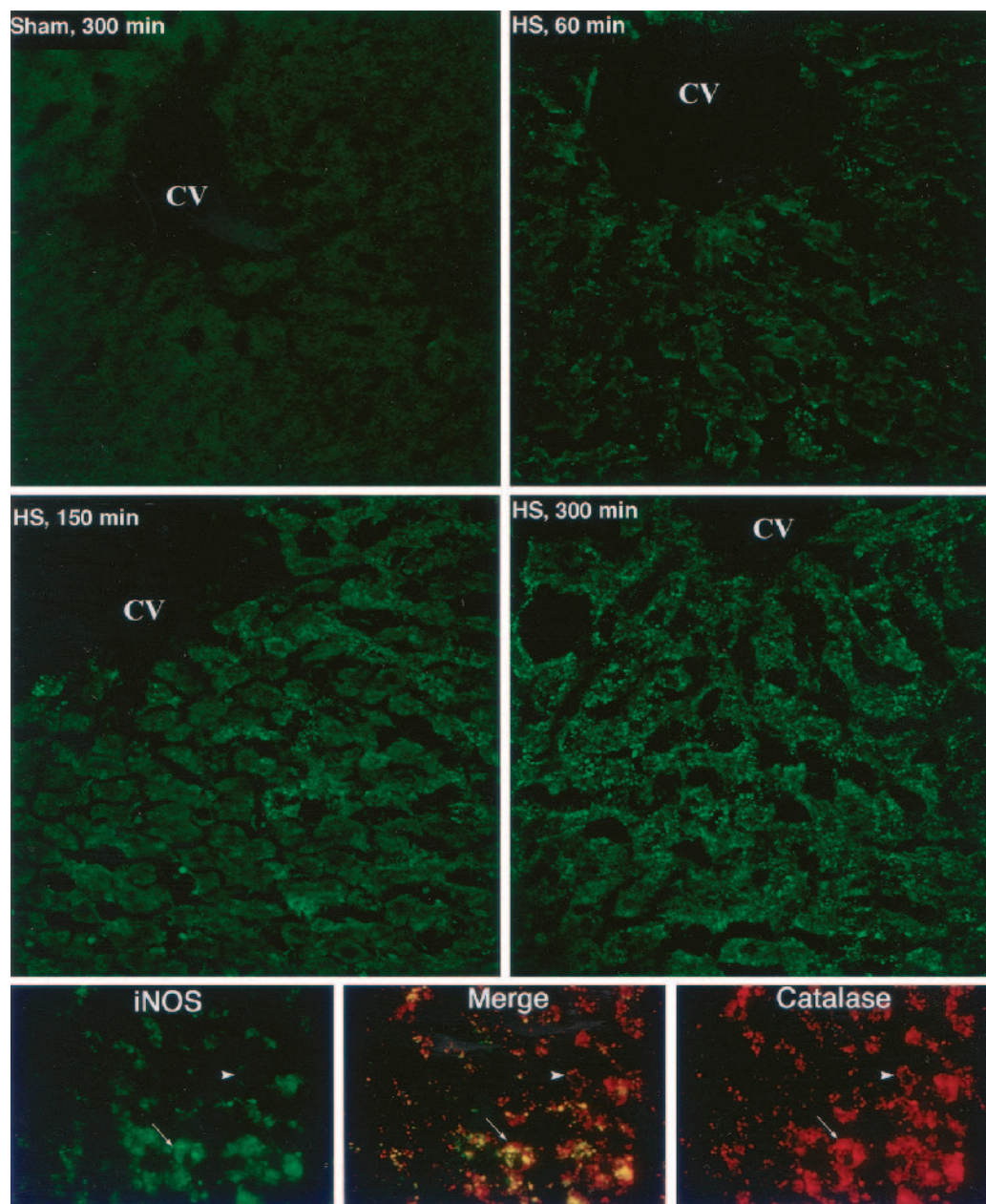


FIG. 2. Immunofluorescence detection of iNOS in rat liver tissues. The expression of iNOS is observed as punctate labeling (green) with increasing frequency as the time of HS increases from 60 to 300 min. Labeling appears to be localized to areas around the central vein (CV) within the liver lobule. Sham-treated control (300 min) animals do not exhibit staining for iNOS. Bottom three panels: Colocalization (yellow) of iNOS (green) with catalase (red) after 300 min of HS. Essentially all iNOS was colocalized with catalase expression as shown by arrow, but not all catalase expression is observed with iNOS (arrow-head), indicating that hepatocytes respond heterogeneously to HS. Bars represent 20 μ m. Immunostaining is representative of three different animals.

several aspects of the inflammatory response following resuscitation from shock are dependent on iNOS-derived NO. Our studies also indicated that hepatic iNOS localizes primarily within peroxisomes of hepatocytes. Our studies in humans confirm that trauma-induced hemorrhage is sufficient to lead to the expression of iNOS mRNA in humans, providing some insight into the clinical relevance of the rodent observations.

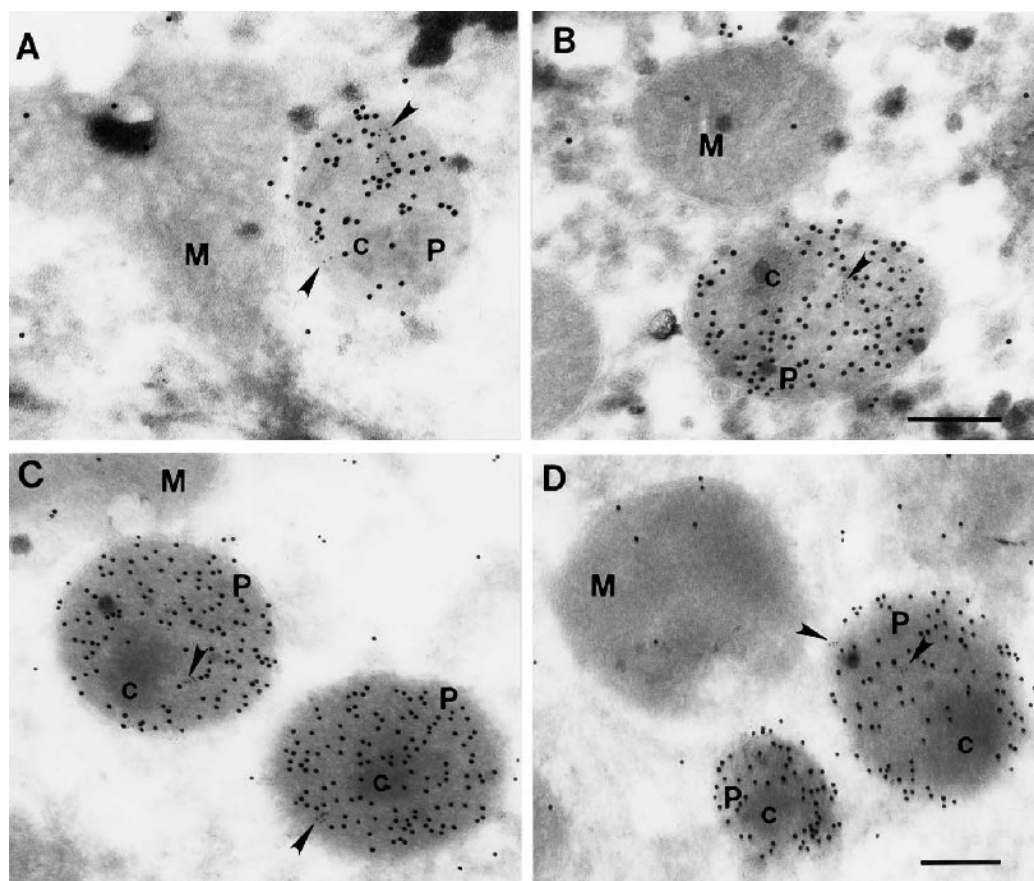
Detection of iNOS by immunohistochemistry in HS has not been described previously. Our immunohistochemical analysis confirmed the increased expression of iNOS protein in the livers of hemorrhaged rats by 1 h of shock and also demonstrated distinct cellular localization of this protein. It appears that iNOS is mainly contained within the hepatocyte, and the pattern of staining is heterogeneous, with some hepatocytes demonstrating no iNOS staining, whereas others stain intensely. However, it is evident from the low magnification views that iNOS is most concentrated in the centrilobular

regions, immediately surrounding the central veins. We would speculate that hypoxia is involved in the upregulation of iNOS because the central lobular region would be the most hypoxic within the hepatic lobule. Previous studies have suggested that hypoxia can contribute to the induction of iNOS (15–18). Such a role of hypoxia is likely to be additional to that of inflammatory mediators known to be released in the process of hemorrhagic shock (1, 2).

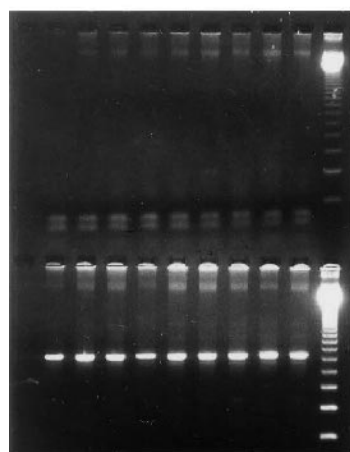
Our immunohistochemical and immunoelectron microscopy analysis revealed that iNOS colocalizes with catalase. As previously mentioned, not all hepatocytes contain iNOS, but where iNOS is present, catalase is present also. The implications of this finding are also unclear, but represent a fascinating area of potential investigation. We have previously shown that iNOS localizes to peroxisomes in hepatocytes treated with a cytokine mixture TNF- α , IFN- γ , and IL-1, a condition that mimics septic shock *in vitro* (11). Other reports of iNOS

FIG. 3. iNOS immunoelectron microscopy in livers of rats subjected to hemorrhagic shock.

Rats were subjected HS or sham procedure. Liver sections were immunostained as described in "Materials and Methods". iNOS (arrowheads, small particles, 5 nm) staining is observed within with the matrix of the peroxisomes (P), not the crystalline core (C). Catalase (large gold particles, 10 nm) expression identifies the organelle as a peroxisome. Also of note is that iNOS staining is not evident in mitochondria (M). A, 2.5 h HS; B and C, 3.5 h HS; D, 5 h HS. Bar for B representative of A and B; bar in D is representative of C and D. Bar = 200 nm. Immunostaining is representative of three different animals.



9 Control Patients



← iNOS →
(316 bp)

← β-actin →
(622 bp)

18 Trauma Patients

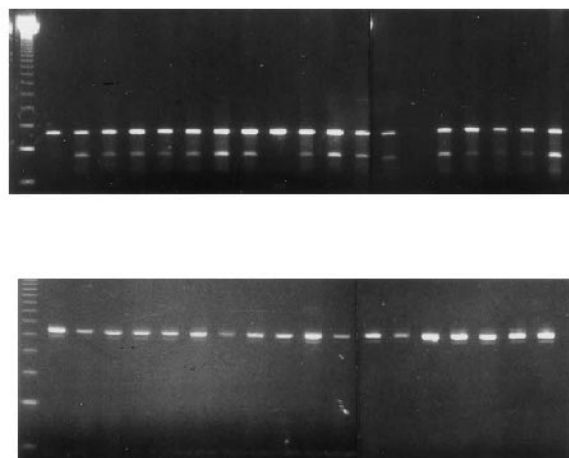


FIG. 4. RT-PCR analysis of iNOS mRNA expression in livers of hypotensive patients with trauma. RNA extracted from liver biopsy specimens was PCR amplified with primers yielding the following size products: iNOS, 316 bp; β-actin, 622 bp. Representative gel of samples from a total of 18 hypotensive patients with trauma and nine patients without trauma.

expression in various cell types under different conditions have localized iNOS to various subcellular regions and structures, including the cytosol (19–23), cytoplasmic vesicles and endoplasmic reticulum (22), Golgi-like membranes (21), and novel iNOS-bearing vesicles (24). In our current study of liver tissue in a rodent model of HS, no iNOS staining was noted in cellular organelles other than peroxisomes, including the plasma membrane, nucleus, nuclear lamina, endoplasmic reticulum, or

mitochondria. It is not surprising that iNOS is localized in peroxisomes, given the primary function of this organelle. Peroxisomes exist in all known mammalian cells with the exception of red blood cells, and one of their primary functions is to contain various antioxidant enzymes, one of which is catalase. The fact that iNOS and catalase coexist suggests that perhaps the peroxisome and its enzymes act to limit the damaging oxidative effects of iNOS-derived NO (25). We are

currently characterizing peroxisomal iNOS in detail, and have demonstrated preliminarily that it is enzymatically active (P. Loughran, D. Stolz, S. Watkins, Y. Vodovotz, and T.R. Billiar, unpublished observations).

Our studies do not provide direct evidence for the mechanism of iNOS upregulation. There has been extensive research in the repletion of iNOS in hepatocytes in other settings, and it is known that high level of IL-1 (26) or a combination of cytokines (IL-1, TNF- α , and IFN- γ) and LPS (27, 28) can induce iNOS. It is possible that TNF- α could reach high local concentrations in the liver within 60 min of shock, and this may synergize with hypoxia. As mentioned above, this is consistent with the early centrilobular expression of iNOS. However, we have been unable to detect iNOS upregulation in cultured hepatocytes exposed to hypoxia alone (data not shown). Therefore, other critical signals are also likely to be important. Importantly, we find that iNOS mRNA levels at 60 min are not different in sham and HS treatments. Previous studies have shown that the *iNOS* gene could be transcribed, but transcripts not accumulate in the steady state, and also not be translated to iNOS protein (28–30).

It is clear from this study and our previous work (5, 13, 14) that HS causes the rapid expression of functionally important levels of iNOS in hepatocytes, and contrary to initial views, neither major inflammatory nor septic stimuli are required for hepatocytes to express biologically significant levels of this enzyme.

ACKNOWLEDGMENTS

This study was supported by the National Institutes of Health (grant nos. P50-GM-53789 and HL 32154 and CA76541). The authors would like to thank the excellent technical assistance of Fran Shagas, Mara Grove Sullivan, and Jamie Popovich of the Center for Biologic Imaging.

REFERENCES

- Peitzman AB, Billiar TR, Harbrecht BG, Kelly E, Udekwo AO, Simmons RL: Hemorrhagic shock. *Curr Probl Surg* 32:925–1002, 1995.
- Jarrar D, Chaudry IH, Wang P: Organ dysfunction following hemorrhage and sepsis: mechanisms and therapeutic approaches. *Int J Mol Med* 4:575–583, 1999.
- Thiemermann C, Szabo C, Mitchell JA, Vane JR: Vascular hyporeactivity to vasoconstrictor agents and hemodynamic decompensation in hemorrhagic shock is mediated by nitric oxide. *Proc Natl Acad Sci USA* 90:267–271, 1993.
- Kelly E, Shah NS, Morgan NN, Watkins SC, Peitzman AB, Billiar TR: Physiologic and molecular characterization of the role of nitric oxide in hemorrhagic shock: evidence that type II nitric oxide synthase does not regulate vascular decompensation. *Shock* 7:157–163, 1997.
- Hierholzer C, Harbrecht B, Menezes JM, Kane J, MacMicking J, Nathan CF, Peitzman AB, Billiar TR, Tweardy DJ: Essential role of induced nitric oxide in the initiation of the inflammatory response after hemorrhagic shock. *J Exp Med* 187:917–928, 1998.
- Chomczynski P, Sacchi N: Single-step method of RNA isolation by acid guanidinium thiocyanate-phenol-chloroform extraction. *Anal Biochem* 162:156–159, 1987.
- Hierholzer C, Kelly E, Tsukada K, Loeffert E, Watkins S, Billiar TR, Tweardy DJ: Hemorrhagic shock induces G-CSF expression in bronchial epithelium. *Am J Physiol* 273:L1058–L1064, 1997.
- Godfrey TE, Kim S-H, Chavira M, Ruff DW, Gray JW, Jensen RH: Quantitative mRNA expression analysis from formalin-fixed, paraffin-embedded tissues using 5' nuclease quantitative RT-PCR. *J Mol Diagn* 2:84–91, 2000.
- Tassone F, Hagerman RJ, Taylor AK, Gane LW, Godfrey TE, Hagerman PJ: Elevated levels of FMR1 mRNA in carrier males: a new mechanism of involvement in the fragile-X syndrome. *Am J Hum Genet* 66:6–15, 2000.
- Godfrey TE, Raja S, Finkelstein SD, Gooding WE, Kelly LA, Luketich JD: Prognostic value of quantitative reverse transcription-polymerase chain reaction in lymph node-negative esophageal cancer patients. *Clin Cancer Res* 7:4041–4048, 2001.
- Stolz DB, Zamora R, Vodovotz Y, Loughran P, Billiar TR, Kim Y-M, Simmons RL, Watkins SC: Peroxisomal localization of inducible nitric oxide synthase in hepatocytes. *Hepatology* 17:81–93, 2002.
- Rajnik M, Salkowski CA, Thomas KE, Li YY, Rollwagen FM, Vogel SN: Induction of early inflammatory gene expression in a murine model of nonre-suscitated, fixed-volume hemorrhage. *Shock* 17:322–328, 2002.
- Menezes J, Hierholzer C, Watkins SC, Lyons V, Peitzman AB, Billiar TR, Tweardy DJ, Harbrecht BG: A novel nitric oxide scavenger decreases liver injury and improves survival after hemorrhagic shock. *Am J Physiol* 277:G144–G151, 1999.
- Menezes JM, Hierholzer C, Watkins SC, Billiar TR, Peitzman AB, Harbrecht BG: The modulation of hepatic injury and heat shock expression by inhibition of inducible nitric oxide synthase after hemorrhagic shock. *Shock* 17:13–18, 2002.
- Melillo G, Musso T, Sica A, Taylor LS, Cox GW, Varesio L: A hypoxia-responsive element mediates a novel pathway of activation of the inducible nitric oxide synthase promoter. *J Exp Med* 182:1683–1693, 1995.
- Xu XP, Pollock JS, Tanner MA, Myers PR: Hypoxia activates nitric oxide synthase and stimulates nitric oxide production in porcine coronary resistance arteriolar endothelial cells. *Cardiovasc Res* 30:841–847, 1995.
- Manukhina EB, Malyshev IY, Smirin BV, Mashina SY, Saltykova VA, Vanin AF: Production and storage of nitric oxide in adaptation to hypoxia. *Nitric Oxide* 3:393–401, 1999.
- Hierholzer C, Harbrecht BG, Billiar TR, Tweardy DJ: Hypoxia-inducible factor-1 activation and cyclo-oxygenase-2 induction are early reperfusion-independent inflammatory events in hemorrhagic shock. *Arch Orthop Trauma Surg* 121:219–222, 2001.
- Stuehr DJ, Marletta MA: Synthesis of nitrite and nitrate in murine macrophage cell lines. *Cancer Res* 47:5590–5594, 1987.
- Förstermann U, Schmidt HHW, Kohlhaas KL, Murad F: Induced RAW 264.7 macrophages express soluble and particulate nitric oxide synthase: inhibition by transforming growth factor- β . *Eur J Pharmacol* 225:161–165, 1992.
- Liu J, Zhao ML, Brosnan CF, Lee SC: Expression of type II nitric oxide synthase in primary human astrocytes and microglia: role of IL-1 β and IL-1 receptor antagonist. *J Immunol* 157:3569–3576, 1996.
- Ishiwata T, Guo F, Naito Z, Asano G, Nishigaki R: Differential distribution of eNOS and iNOS mRNA in rat heart after endotoxin administration. *Jpn Heart J* 38:445–455, 1997.
- Htain WW, Leong SK, Ling EA: In vivo expression of inducible nitric oxide synthase in supraventricular amoeboid microglial cells in neonatal BALB/c and athymic mice. *Neurosci Lett* 223:53–56, 1997.
- Vodovotz Y, Russell D, Xie Q-W, Bogdan C, Nathan C: Vesicle association of nitric oxide synthase from primary mouse macrophages. *J Immunol* 154:2914–2925, 1995.
- Pacelli R, Wink DA, Cook JA, Krishna MC, DeGraff W, Friedman N, Tsokos M, Samuni A, Mitchell JB: Nitric oxide potentiates hydrogen peroxide-induced killing of *Escherichia coli*. *J Exp Med* 182:1469–1479, 1995.
- Geller DA, de Vera ME, Russell DA, Shapiro RA, Nussler AK, Simmons RL, Billiar TR: A central role for IL-1 β in the in vitro and in vivo regulation of hepatic inducible nitric oxide synthase: IL-1 β induces hepatic nitric oxide synthesis. *J Immunol* 155:4890–4898, 1995.
- Curran RD, Billiar TR, Stuehr DJ, Hoffmann K, Simmons RL: Hepatocytes produce nitrogen oxides from L-arginine in response to inflammatory products of Kupffer cells. *J Exp Med* 170:1769–1774, 1989.
- Geller DA, Nussler AK, Di Silvio M, Lowenstein CJ, Shapiro RA, Wang SC, Simmons RL, Billiar TR: Cytokines, endotoxin, and glucocorticoids regulate the expression of inducible nitric oxide synthase in hepatocytes. *Proc Natl Acad Sci USA* 90:522–526, 1993.
- Vodovotz Y, Bogdan C, Paik J, Xie Q-W, Nathan C: Mechanisms of suppression of macrophage nitric oxide release by transforming growth factor- β . *J Exp Med* 178:605–613, 1993.
- Taylor BS, Geller DA: Molecular regulation of the human inducible nitric oxide synthase (iNOS) gene. *Shock* 13:413–424, 2000.